ZHENYU (ZACH) WANG

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RESEARCH INTERESTS

My research lies at the intersection of statistics, optimization, machine learning, and artificial intelligence. I develop statistical methodology and theory for modern data-scientific problems, with an emphasis on multi-source robust learning, causal inference, statistical uncertainty quantification, and scalable learning algorithms. My recent work studies generative AI and large language models from a statistical and optimization perspective, focusing on efficient inference-time computation and reinforcement learning for LLM post-training.

EDUCATION

Rutgers University Sept. 2022 – May 2027 (expected)
Ph.D. in Statistics
Advisors: Zijian Guo and Yifan Hu

Columbia University Sept. 2019 – May 2021
M.S. in Applied Analytics

Shanghai Jiao Tong University Sept. 2014 – June 2019
Bachelor's degree in Finance

PUBLICATIONS

* indicates equal contribution.

- **Distributionally Robust Learning for Multisource Unsupervised Domain Adaptation**
  Zhenyu Wang, Peter Bühlmann, Zijian Guo
  The Annals of Statistics, 54(2): 570–596, 2026.
  Develops robust prediction methods that combine heterogeneous data sources for the unknown target domain.

- **SIHR: Statistical Inference in High-Dimensional Linear and Logistic Regression Models**
  Prabrisha Rakshit*, **Zhenyu Wang***, Tony Cai, Zijian Guo
  The R Journal, 16(3): 27–45, 2024.
  Provides methods and software for confidence intervals and hypothesis testing in high-dimensional regression.

PREPRINTS AND SUBMITTED MANUSCRIPTS

Conferences

- **Pareto Frontier of LLM Test-Time Compute: Adaptive Rollouts and Voting Under Token Budgets**
  Zhenyu Wang, Xiaozhi Zhu, Yifan Hu
  Under review at *ICLR 2027*.
  Accepted at *NeurIPS 2026 workshops*: **FLLMPT (oral)**, MLxOR, RAAAI, and TTCL.
  Characterizes the cost-accuracy frontier of LLM inference theoretically and develops budget-aware inference strategies.

- **Dr. OPD: Learning What to Follow for Optimal On-Policy Distillation of Large Language Models**
  Zhenyu Wang*, Tianze Wang*, Linjun Zhang, Yifan Hu
  Under review at *ICLR 2027*.
  Formulates optimal on-policy distillation as a bilevel optimization, adaptively weighting dense token-wise teacher feedback to improve student task performance.

Journals

- **Statistical Analysis of Conditional Group Distributionally Robust Optimization with Cross-Entropy Loss**
  Zijian Guo, **Zhenyu Wang**, Yifan Hu, Francis Bach
  Under review at *The Annals of Statistics*.
  Develops robust classifiers and uniformly valid statistical inference for multi-source domain adaptation.

- **StablePCA: Distributionally Robust Learning of Shared Representations from Multi-Source Data**
  Zhenyu Wang, Molei Liu, Jing Lei, Francis Bach, Zijian Guo
  Extracts shared low-dimensional representations from heterogeneous sources via distributionally robust optimization, preserving biological structure while suppressing batch-specific artifacts.

- **Causal Invariance Learning via Efficient Nonconvex Optimization**
  Zhenyu Wang[*], Yifan Hu[*], Peter Bühlmann, Zijian Guo
  Formulates causal learning as a nonconvex optimization problem, and develops an efficient algorithm that solves the nonconvex problem to global optimality in polynomial time.

AWARDS

- ETH Zürich Young Researcher Travel Award — July 2025
- IMS ICSDS Student Travel Award — Dec. 2024
- Rutgers University Research Fellowship — July 2024
- Winner, Cohen Student Research Poster Competition — March 2024

PRESENTATIONS

- NeurIPS / Annals of Statistics collaboration — invited presentation; San Diego, CA — Dec. 2025
- INFORMS Annual Meeting; Atlanta, GA — Oct. 2025
- Joint Statistical Meetings (JSM); Nashville, TN — Aug. 2025
- New England Statistics Symposium (NESS), Yale University; New Haven, CT — May 2025
- IMS International Conference on Statistics and Data Science (ICSDS); Nice, France — Dec. 2024
- International Conference on Frontiers of Data Science; Hangzhou, China — Aug. 2024
- Joint Conference on Statistics and Data Science in China; Kunming, Yunnan, China — July 2024
- New England Statistics Symposium (NESS), University of Connecticut; Storrs, CT — May 2024
- International Conference on Frontiers of Data Science; Hangzhou, China — Aug. 2023

TEACHING EXPERIENCE

Teaching Assistant, Rutgers University

- Statistical Learning for Data Science — Spring & Fall 2026
- Data Mining — Spring 2025
- Data Wrangling with R — Spring & Fall 2024
- Recitation for the Statistics Ph.D. Qualifying Examination — Summer 2024
- Statistical Inference (Ph.D. core course) — Spring 2024
- Theory of Statistics (Ph.D. core course) — Spring 2024

Invited Guest Lectures, Rutgers University

- Data Mining — Spring 2025 & Fall 2026

ACADEMIC SERVICE

Conference Review

- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- Conference on Neural Information Processing Systems (NeurIPS)

Journal Review

- Journal of the Royal Statistical Society, Series B: Statistical Methodology (JRSSB)
- Journal of the American Statistical Association (JASA)
- Bernoulli
- TEST
- The American Statistician
- Scientia Sinica Mathematica